
Midterm Project Report: Depth Any Panoramas (DAP)

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Abstract

This midterm report outlines the progress and revised methodology for the “Depth Any Panoramas” (DAP) project. We aim to establish a foundation model for panoramic metric depth estimation that generalizes across diverse indoor and outdoor scenes. Addressing the limitations of current state-of-the-art methods, which often fail in complex outdoor environments due to domain gaps, our approach leverages a data-in-the-loop paradigm. We utilize a massive dataset of 2 million synthetic and real-world panoramas, constructed via a novel engine integrating AirSim360 and Qwen2-VL filtering. We introduce a three-stage training pipeline to bridge domain gaps and a novel model architecture incorporating a range mask head and distortion-aware losses. This report details the data engine construction, the mathematical formulation of our loss functions, implementation details, and a theoretical discussion of the proposed method.

1 Revised Project Proposal

1.1 Problem Statement

Panoramic depth estimation is a cornerstone of spatial intelligence, offering a complete $360^\circ \times 180^\circ$ field of view that is essential for applications such as autonomous driving, robotic navigation, and Virtual Reality (VR) (3). Unlike perspective images, panoramas capture the entire surrounding environment, providing holistic geometric information.

However, estimating accurate metric depth from a single panoramic image remains a significant challenge. Existing methods face two primary bottlenecks:

1. **The Domain Gap:** There is a severe scarcity of high-quality, dense depth annotations for real-world panoramic scenes. Models trained solely on synthetic data (e.g., Structured3D) often fail to generalize to real-world textures and lighting. Conversely, models trained on limited real-world data lack the diversity required for robust performance.
2. **Outdoor Generalization:** Most current datasets are confined to indoor environments. Extending depth estimation to outdoor scenes is difficult due to infinite depth regions (e.g., sky) and complex, unstructured geometries. State-of-the-art methods often produce distorted or collapsed predictions in these areas.

1.2 Project Objectives

We propose **DAP (Depth Any Panoramas)**, a foundation model designed to achieve robust, metrically consistent depth estimation. Our revised objectives for this milestone include:

- **Data Engine Scaling:** Construct the largest panoramic depth dataset to date (~2 million samples), merging high-quality synthetic data from Structured3D and AirSim360 with filtered real-world unlabeled data to enable scalable semi-supervised learning (1; 2).
- **Pipeline Innovation:** Implement a three-stage training pipeline (Scene-Invariant Labeler → Realism-Invariant Labeler → DAP) that progressively utilizes pseudo-labels to bridge the synthetic-to-real domain gap.
- **Geometric Consistency:** Develop a network architecture using a DINOv3-Large backbone (5), augmented with a plug-and-play range mask head and distortion-aware losses to handle the unique challenges of Equirectangular Projection (ERP).

2 Related Work

2.1 Perspective Depth Estimation

Perspective depth estimation has advanced rapidly due to the availability of large-scale datasets. Recent metric and scale-invariant models, such as Metric3D and Depth Anywhere, have demonstrated impressive zero-shot generalization capabilities. Some approaches fine-tune large generative models like Stable Diffusion on high-quality data to achieve dense predictions. However, these perspective paradigms are inherently restricted by a limited Field of View (FoV) and fail to capture the global 360° spatial geometry required for omnidirectional perception (4).

2.2 In-Domain Panoramic Depth

Early panoramic methods focused on in-domain settings, where models are trained and evaluated on the same dataset. To address the severe distortion introduced by Equirectangular Projection (ERP), researchers developed distortion-aware convolution filters and projection-driven strategies (e.g., Cubemap projection). While effective within their specific domains, these methods often suffer from overfitting and lack the generalization ability needed for "in-the-wild" applications.

2.3 Zero-Shot and Cross-Domain Approaches

To overcome data limitations, recent work has shifted towards zero-shot estimation.

- **Knowledge Distillation:** Methods like PanDA and Depth Anywhere leverage pre-trained perspective models to generate pseudo-labels for panoramas. While they improve generalization, they typically output relative depth, which lacks the metric scale necessary for robotics (6).
- **Metric Frameworks:** UniK3D reformulates depth in spherical coordinates using harmonic ray representations to enhance wide-FoV generalization (3). Similarly, Depth Any Camera (DAC) attempts to unify diverse imaging geometries (4).

Despite these advances, these metric frameworks still struggle with diverse real-world scenes, particularly outdoors, due to the limited scale of training data. Our work differentiates itself by explicitly constructing a massive hybrid data engine to solve this data scaling issue.

3 Detailed Description of Methods

Our method unifies metric estimation across domains through three synergistic components: a large-scale data engine, a progressive training pipeline, and a geometry-aware network design.

3.1 Data Engine Construction: DAP-2M

We constructed **DAP-2M**, a dataset comprising approximately 2 million samples. This engine is composed of:



Figure 1: Overview of the proposed progressive three-stage pipeline. Stage 1 trains a **Scene-Invariant Labeler** on high-quality synthetic indoor and outdoor data. Stage 2 introduces a **Realism-Invariant Labeler**, using a PatchGAN-based discriminator to select high-confidence pseudo-labeled samples from real data. Stage 3 performs final **DAP** training on the combined dataset.

3.1.1 Labeled Synthetic Data (110k)

To provide accurate ground truth, we utilize:

- **Indoor:** 20k samples from the Structured3D dataset, providing high-quality indoor scene layouts (1).
- **Outdoor (AirSim360):** Addressing the lack of outdoor supervision, we rendered 90k samples using **AirSim360**, a simulator based on Unreal Engine 5. We simulated drone flights with diverse mid- and low-altitude trajectories to capture realistic lighting, weather conditions, and geometries across 5 representative environments (e.g., New York City, Rome) (2).

3.1.2 Unlabeled Real Data (1.9M)

We collected over 1.7 million panoramic images from the internet and video frames. To ensure data quality and balance:

- **Filtration:** We employed the large multimodal model **Qwen2-VL** to automatically categorize panoramas into indoor and outdoor scenes, yielding 250k indoor and 1.45M outdoor samples (7).
- **Generation:** Since indoor panoramas were relatively scarce, we supplemented this domain by generating 200k synthetic indoor samples using the panoramic generation model **DiT360**.

3.2 Three-Stage Training Pipeline

As illustrated in **Figure 1**, we employ a semi-supervised pipeline:

1. **Stage 1 (Scene-Invariant Labeler):** We train an initial model on the 110k labeled synthetic samples. This model learns physically grounded depth cues that generalize across scene layouts without overfitting to specific textures.
2. **Stage 2 (Realism-Invariant Labeler):** To bridge the synthetic-to-real gap, we employ a **PatchGAN-based discriminator**. This discriminator is trained to distinguish between synthetic ground truth (labeled as "real") and the Stage 1 model's predictions on real data (labeled as "fake"). We then select the top 600k high-confidence samples (300k indoor, 300k outdoor) based on the discriminator's score. These samples are used to train a second labeler robust to realistic appearance variations.
3. **Stage 3 (DAP Training):** The final foundation model is trained on the union of labeled synthetic data and the refined pseudo-labeled real data (approx 1.9M total).

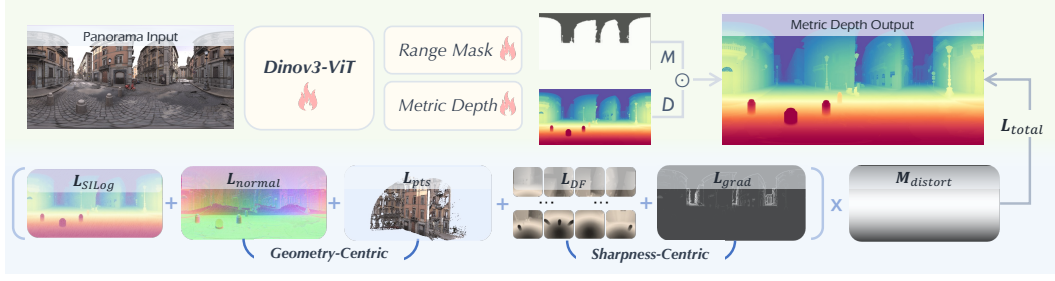


Figure 2: Architecture of the proposed **DAP network**. Built upon DINOv3-Large (5), it features a distortion-aware depth decoder and a range mask head. Training is guided by multi-level losses: $\mathcal{L}_{DF}$ for sharpness, $\mathcal{L}_{grad}$ for edges, and $\mathcal{L}_{normal}/\mathcal{L}_{pts}$ for geometric consistency.

3.3 Model Architecture

The DAP network (**Figure 2**) utilizes **DINOv3-Large** as a visual encoder to leverage strong pre-trained visual priors (5). The decoder consists of two primary heads.

3.3.1 Range Mask Head

To adapt to diverse spatial scales, we introduce a plug-and-play range mask head. The network predicts a binary mask M defining valid spatial regions for distance thresholds $\tau \in \{10, 20, 50, 100\}$ meters. This effectively filters out unreliable far-field predictions in complex outdoor scenes. The optimization objective is:

$$\mathcal{L}_{mask} = \|M - M_{gt}\|^2 + 0.5\mathcal{L}_{Dice}(M, M_{gt}) \quad (1)$$

where M_{gt} is the ground truth mask derived from depth thresholds.

3.3.2 Metric Depth Head and Loss Functions

We employ a composite loss function to ensure both pixel-wise accuracy and structural fidelity.

Sharpness-Centric Optimization: To mitigate the distortion artifacts inherent to ERP, particularly at the poles, we utilize a **Dense Fidelity loss** ($\mathcal{L}_{DF}$). We decompose the panorama into $N = 12$ perspective patches (virtual cameras positioned at the vertices of an icosahedron). We compute the Gram-based similarity on these distortion-free patches:

$$\mathcal{L}_{DF} = \frac{1}{N} \sum_{k=1}^N \|D_{pred}^{(k)} \odot D_{pred}^{(k)\top} - D_{gt}^{(k)} \odot D_{gt}^{(k)\top}\|_F^2 \quad (2)$$

Additionally, a **Gradient Loss** ($\mathcal{L}_{grad}$) is applied to high-frequency edge regions defined by a Sobel-based edge mask M_E :

$$\mathcal{L}_{grad} = \mathcal{L}_{SILog}(M_E \odot D_{pred}, M_E \odot D_{gt}) \quad (3)$$

Geometry-Centric Optimization: To ensure 3D consistency, we incorporate surface normal and point cloud losses. The normal loss $\mathcal{L}_{normal}$ minimizes the L1 distance between predicted and ground truth surface normal fields. Similarly, the point cloud loss $\mathcal{L}_{pts}$ minimizes the distance between back-projected 3D points:

$$\mathcal{L}_{normal} = \|n_{pred} - n_{gt}\|_1, \quad \mathcal{L}_{pts} = \|P_{pred} - P_{gt}\|_1 \quad (4)$$

Total Objective: We apply a **Distortion Map** $M_{distort}$ to the final loss to account for the non-uniform pixel density of ERP (where pixels near the poles cover smaller solid angles). The total objective is:

$$\mathcal{L}_{total} = M_{distort} \odot (\lambda_1 \mathcal{L}_{SILog} + \lambda_2 \mathcal{L}_{DF} + \lambda_3 \mathcal{L}_{grad} + \lambda_4 \mathcal{L}_{normal} + \lambda_5 \mathcal{L}_{pts} + \lambda_6 \mathcal{L}_{mask}) \quad (5)$$

where the weights are set as $\lambda_1 = 1.0, \lambda_2 = 0.4, \lambda_3 = 5.0, \lambda_4 = 2.0, \lambda_5 = 2.0, \lambda_6 = 2.0$.

3.4 Implementation Details

The model is implemented using PyTorch. Training is conducted on H20 GPUs. We use the Adam optimizer with a learning rate of $5e^{-6}$ for the ViT backbone and $5e^{-5}$ for the decoders. The input resolution is set to 512×1024 . Data augmentation strategies include color jittering, horizontal translation, and flipping to enhance robustness.

4 Discussion of Proposed Method

4.1 Novelty

The DAP project introduces three primary innovations:

1. **Scale:** We present the first panoramic depth dataset exceeding 2 million samples. This scale significantly surpasses previous benchmarks like DAC (800k) and UniK3D (694k), enabling the training of true foundation models.
2. **Process:** The three-stage filtering pipeline is a novel application of semi-supervised learning specifically tailored to the "Sim-to-Real" gap in panoramic vision. By using a discriminator to vet pseudo-labels, we prevent the model from learning artifacts common in synthetic data.
3. **Architecture:** The hybrid optimization strategy—combining perspective-patch consistency ($\mathcal{L}_{DF}$) with global ERP losses—effectively solves the "pole distortion" problem that plagues standard CNNs/ViTs on spherical images.

4.2 Theoretical Analysis & Advantages

The proposed method offers significant theoretical advantages over existing baselines.

- **Metric vs. Relative:** Unlike Depth Anywhere, which provides relative depth (scale-invariant), DAP provides absolute metric depth. This is achieved through the Range Mask head, which enforces scale consistency by explicitly modeling distance thresholds. This is crucial for autonomous agents that need to know if an obstacle is 5 meters or 50 meters away.
- **Outdoor Robustness:** Standard panoramic models often collapse in "infinite depth" regions like the sky. Our inclusion of AirSim360 data and the specific mask loss allows the model to handle these singularities gracefully, maintaining structural integrity where other models fail.

4.3 Limitations

- **Computational Cost:** The reliance on the DINOv3-Large backbone ensures high-quality features but entails significant computational overhead (FLOPs), potentially limiting real-time deployment on resource-constrained edge devices (e.g., mobile robots).
- **Pseudo-Label Quality:** While the discriminator filters low-quality samples, the system ultimately relies on pseudo-labels for real-world training. Any systematic bias in the teacher model (Realism-Invariant Labeler) may propagate to the final DAP model.

References

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